

Module 5 Assignment: Advanced SQL Aggregation and Window Functions
 CPT 166 – Fundamentals of SQL
 Sylvia Schlotterbeck 10/12/25 (Completed final 3 queries - 10/15/25)

1. GROUP BY & HAVING – Orders per Customer

```
SELECT customer_id, COUNT(order_id) AS number_of_orders
FROM orders
GROUP BY customer_id
HAVING COUNT(order_id) >= 5
ORDER BY customer_id;
```

	customer_id integer	number_of_orders bigint
1	1	6
2	3	7
3	4	13
4	5	18
5	6	7
6	7	11
7	9	17
8	10	14
9	11	10
10	12	6
11	14	8
12	15	5
13	17	6
14	19	8
15	20	30
16	21	7
17	23	5
18	24	19
19	25	15
20	27	6
21	28	8
22	29	5
Total rows: 72		Query complete 00:00:00.073

2. GROUP BY with Aggregate – Total Sales per Category

```
SELECT category_id, SUM(quantity * price) AS total_sales
FROM order_details
INNER JOIN products ON order_details.product_id = products.product_id
GROUP BY category_id
ORDER BY total_sales DESC;
```

	category_id integer	total_sales numeric
1	1	309582.25
2	4	269128.30
3	6	190682.69
4	3	190328.54
5	8	149059.53
6	2	122343.00
7	7	111395.00
8	5	106848.00

3. ANY – Products Above a Threshold

```
SELECT product_id, product_name, price
FROM products
WHERE price > ANY (
    SELECT price
    FROM products
    WHERE category_id = 1
);
```

	product_id [PK] integer	product_name character varying (255)	price numeric (10,2)
1	1	Chais	18.00
2	2	Chang	19.00
3	3	Aniseed Syrup	10.00
4	4	Chef Antons Cajun Seasoning	22.00
5	5	Chef Antons Gumbo Mix	21.35
6	6	Grandmas Boysenberry Spread	25.00
7	7	Uncle Bobs Organic Dried Pears	30.00
8	8	Northwoods Cranberry Sauce	40.00
9	9	Mishi Kobe Niku	97.00
10	10	Ikura	31.00
11	11	Queso Cabrales	21.00
12	12	Queso Manchego La Pastora	38.00
13	13	Konbu	6.00
14	14	Tofu	23.25
15	15	Genen Shouyu	15.50
16	16	Pavlova	17.45
17	17	Alice Mutton	39.00
18	18	Carnarvon Tigers	62.50
19	19	Teatime Chocolate Biscuits	9.20
Total rows: 75		Query complete 00:00:00.086	

4. ALL – Products Cheaper than All in a Category

```
SELECT product_id, product_name, price
FROM products
WHERE price < ALL (
    SELECT price
    FROM products
    WHERE category_id = 5
);
```

	product_id [PK] integer 	product_name character varying (255) 	price numeric (10,2) 
1	13	Konbu	6.00
2	24	Guarani Fantastica	4.50
3	33	Geitost	2.50

5. Window Function – Ranking Customers by Orders

```
WITH customer_rank AS (  
  SELECT  
    customer_id,  
    COUNT(order_id) AS number_of_orders  
  FROM orders  
  GROUP BY customer_id)  
  
SELECT  
  customer_id, number_of_orders,  
  DENSE_RANK() OVER (ORDER BY number_of_orders DESC) AS rank  
FROM customer_rank;
```

	customer_id integer	number_of_orders bigint	rank bigint
1	71	31	1
2	20	30	2
3	63	28	3
4	24	19	4
5	37	19	4
6	65	18	5
7	35	18	5
8	5	18	5
9	9	17	6
10	44	15	7
11	25	15	7
12	87	15	7
13	39	14	8
14	41	14	8
15	34	14	8
16	46	14	8
17	89	14	8
Total rows: 89		Query complete 00:00:00.079	

6. Window Function – Running Total of Orders

```
SELECT
    order_id, customer_id, order_date,
    ROW_NUMBER() OVER (ORDER BY order_date ASC) AS
orders_running_total
FROM orders
ORDER BY order_date;
```

	order_id [PK] integer	customer_id integer	order_date date	orders_running_total bigint
1	10248	90	2021-07-04	1
2	10249	81	2021-07-05	2
3	10250	34	2021-07-08	3
4	10251	84	2021-07-08	4
5	10252	76	2021-07-09	5
6	10253	34	2021-07-10	6
7	10254	14	2021-07-11	7
8	10255	68	2021-07-12	8
9	10256	88	2021-07-15	9
10	10257	35	2021-07-16	10
11	10258	20	2021-07-17	11
12	10259	13	2021-07-18	12
13	10260	55	2021-07-19	13
14	10261	61	2021-07-19	14
15	10262	65	2021-07-22	15
16	10263	20	2021-07-23	16
17	10264	24	2021-07-24	17
18	10265	7	2021-07-25	18
19	10266	87	2021-07-26	19
20	10267	25	2021-07-29	20
21	10268	33	2021-07-30	21
Total rows: 830		Query complete 00:00:00.081		

7. Partitioned Average

```
SELECT
    category_name, product_name, price,
    ROUND(AVG(price) OVER (
        PARTITION BY products.category_id),2) AS category_average_price
FROM products
INNER JOIN categories ON products.category_id = categories.category_id;
```

	category_name character varying (255) 🔒	product_name character varying (255) 🔒	price numeric (10,2) 🔒	category_average_price numeric 🔒
1	Beverages	Guarani Fantastica	4.50	37.98
2	Beverages	Ipoh Coffee	46.00	37.98
3	Beverages	Chartreuse verte	18.00	37.98
4	Beverages	Cote de Blaye	263.50	37.98
5	Beverages	Steeleye Stout	18.00	37.98
6	Beverages	Sasquatch Ale	14.00	37.98
7	Beverages	Lakkalikeeri	18.00	37.98
8	Beverages	Rhenbreu Klosterbier	7.75	37.98
9	Beverages	Outback Lager	15.00	37.98
10	Beverages	Chais	18.00	37.98
11	Beverages	Laughing Lumberjack Lager	14.00	37.98
12	Beverages	Chang	19.00	37.98
13	Condiments	Gula Malacca	19.45	23.06
14	Condiments	Original Frankfurter grune Soae	13.00	23.06
15	Condiments	Northwoods Cranberry Sauce	40.00	23.06
16	Condiments	Louisiana Hot Spiced Okra	17.00	23.06
17	Condiments	Genen Shouyu	15.50	23.06
18	Condiments	Grandmas Boysenberry Spread	25.00	23.06
19	Condiments	Louisiana Fiery Hot Pepper Sauce	21.05	23.06
20	Condiments	Vegie-spread	43.90	23.06
21	Condiments	Aniseed Syrup	10.00	23.06
22	Condiments	Chef Antons Cajun Seasoning	22.00	23.06
23	Condiments	Chef Antons Gumbo Mix	21.35	23.06
24	Condiments	Sirop d arable	28.50	23.06
25	Confections	Scottish Longbreads	12.50	25.16
26	Confections	Pavlova	17.45	25.16
27	Confections	Teatime Chocolate Biscuits	9.20	25.16
Total rows: 77		Query complete 00:00:00.055		